Design of Enterprise IoT



Lecture 1

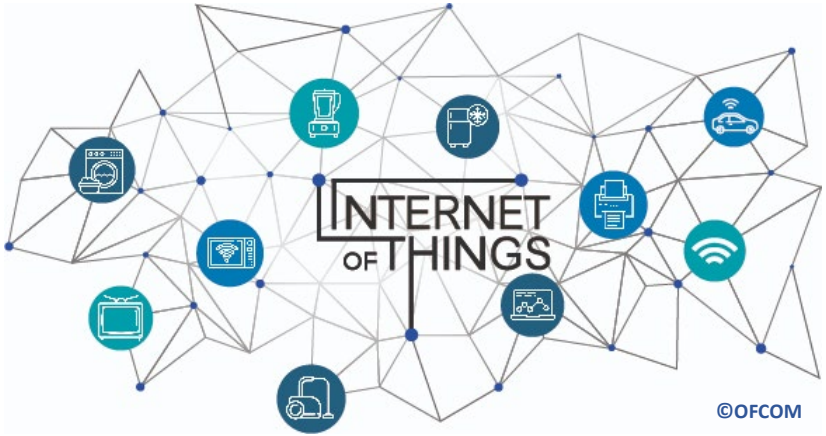
Introduction & Unit Overview

A/Prof Chun Ouyang

About Me

- Associate professor and the Academic Lead in Higher Degree Research (HDR) in the School of Information Systems.
- Program Lead of the JIT-QUT 3+1 Bachelor of IT Joint Program since 2016.
- Taught numerous IT-related subjects in both Bachelor of IT and Master of IT courses since 2009. She also provides supervision and mentoring in various industry-based coursework projects, such as Capstone projects and Work Integrated Learning (WIL) projects.
- Lead the research team on "Explainable Analytics for Machine Intelligence" (xami-lab.org) underpinned by machine learning, explainable Artificial Intelligence, and data analytics.
- Actively seeking candidates for VRES, MYRES, and honours/MPhil/PhD research.
- Check my QUT staff profile for more information

Introduction to
The Internet-of-Things (IoT)



IoT Essentials

- A network of physical elements empowered by:



Sensors | Identifiers | Software | Internet Connectivity

- The network of things, with unique **identification**, embedded with **software** intelligence, **sensors**, and ubiquitous **connectivity to the Internet**.

IoT Components

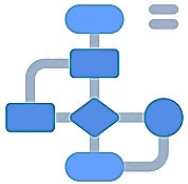
- IoT connects four components:



People



Data



Process



Things

- **Processes** allowing **things** to be connected over the **Internet** to exchange **data**
- Internet of “Everything”

IoT Definitions

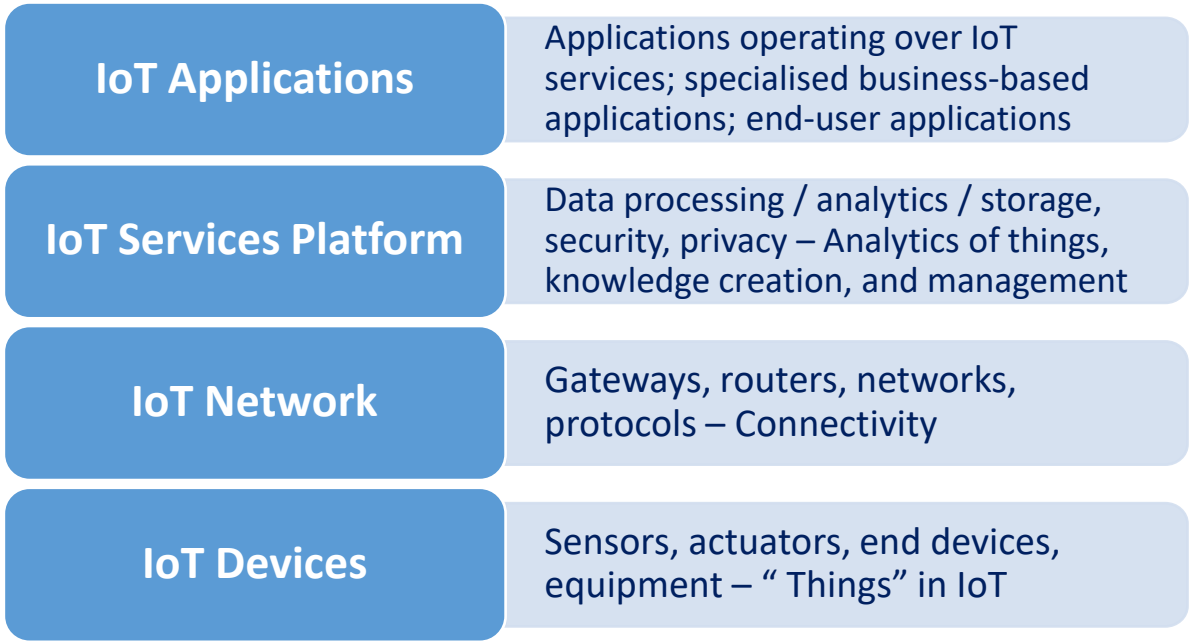
"IoT is the **network** of **things**, with **device identification, embedded intelligence,** and **sensing and acting capabilities**, connecting **people** and **things** over the **Internet**."

(Rayes and Salam, 2022)

"The internet of things, or IoT, is a system of **interrelated computing devices, mechanical and digital machines, objects, animals or people** that are provided with unique identifiers (**UIDs**) and the ability to transfer **data** over a **network** *without requiring human-to-human or human-to-computer interaction*."

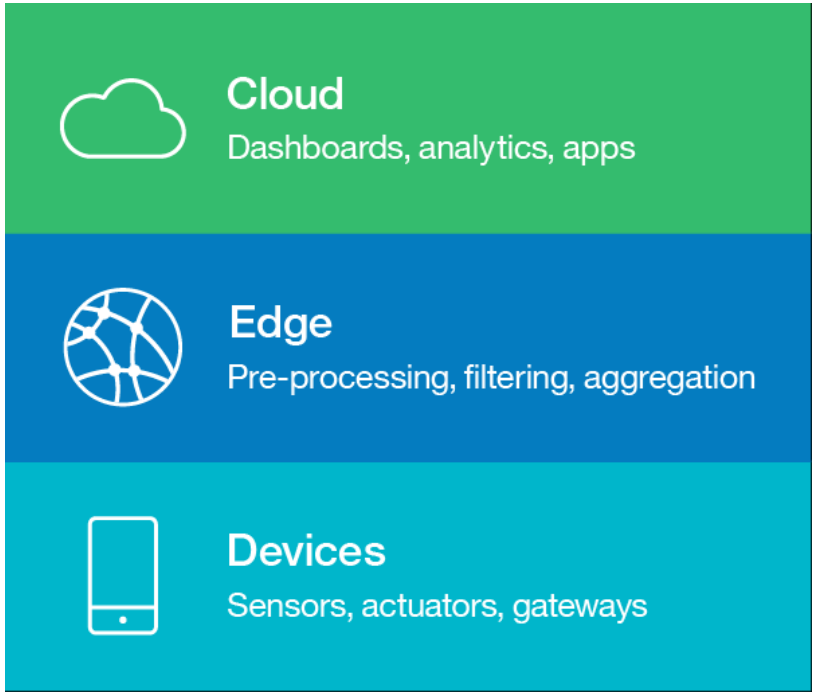
(from Wikipedia)

IoT Reference Framework



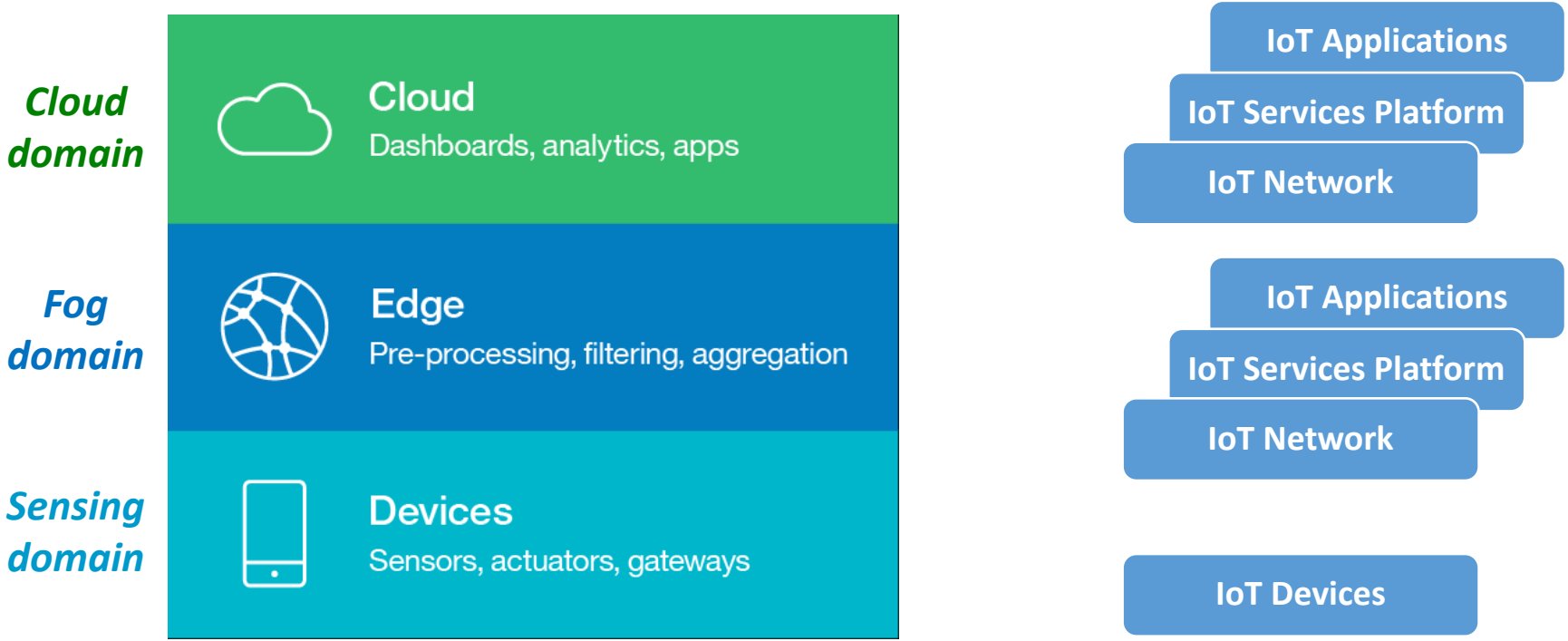
(Rayes and Salam, 2022)

IoT Layered Architecture



- IoT Applications
- IoT Services Platform
- IoT Network
- IoT Devices

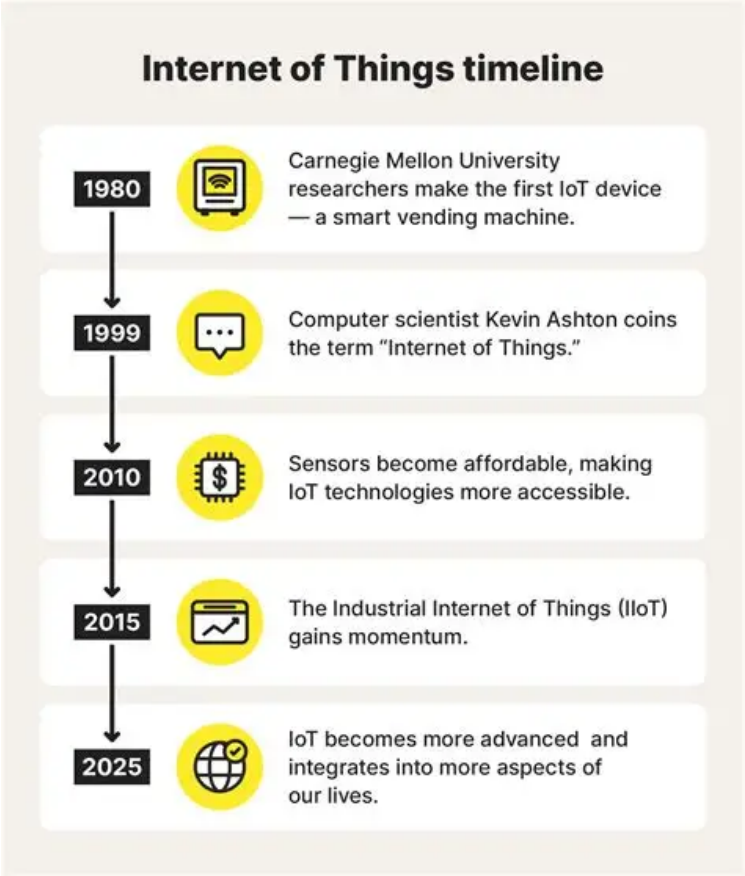
IoT Layered Architecture



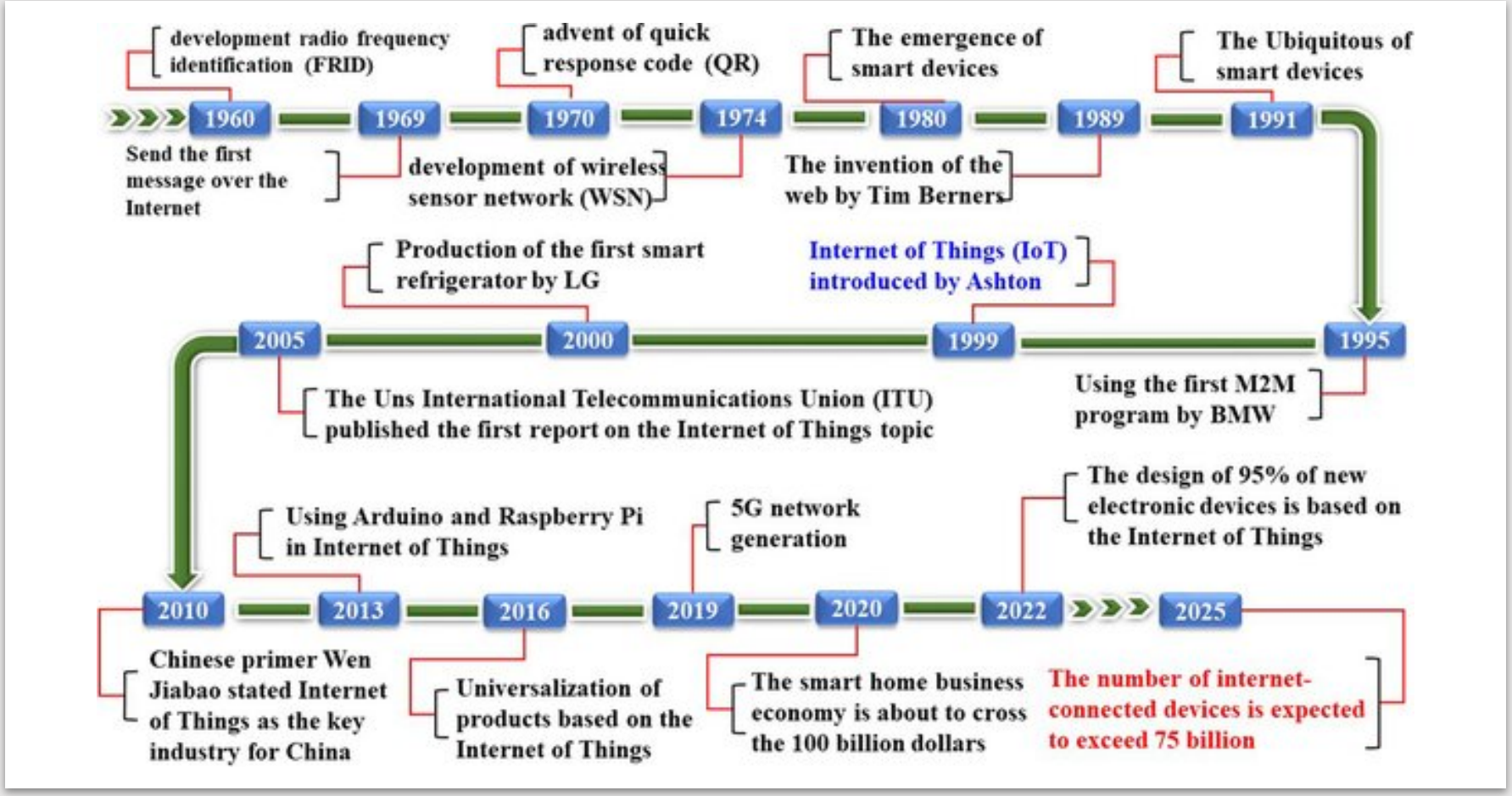


IoT in Practice

Evolution of IoT Technologies



(us.norton.com)



Amirian et al., Seamless integration of Internet of Things, miniaturization, and environmental chemical surveillance. Environ Monit Assess 196, 582 (2024).
DOI: 10.1007/s10661-024-12698-9

IoT Use Scenarios



Wearables:
Offer tracking on personalized data such as daily steps, heart rate, calorie burned that result in improving one's health and fitness.

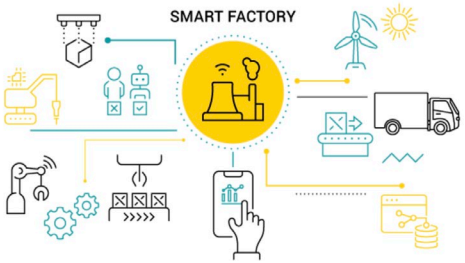


Nest:
Smart thermostat and security system to control temperature and monitor home through a smartphone

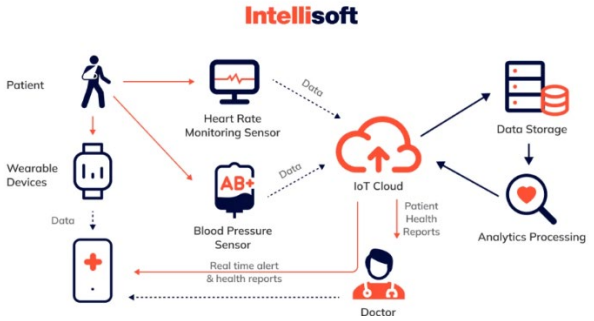


Connected car:
Optimising vehicles overall functions

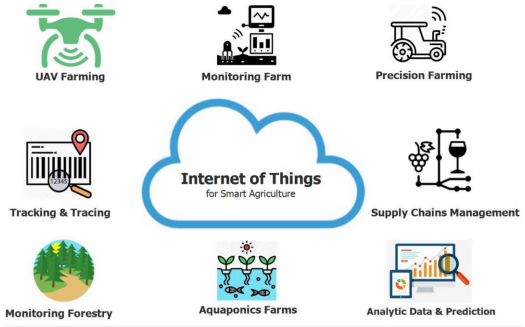
Enterprise IoT Applications



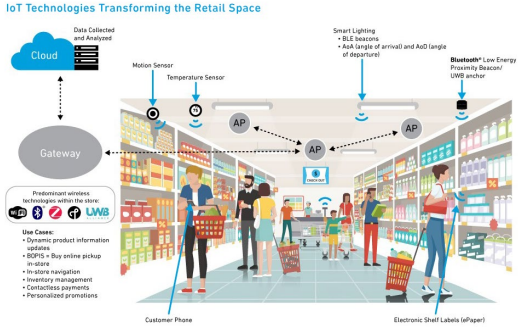
Source: AVSystem, 2024



Source: Intellisoft, 2024



(Quy et al., 2022)



Source: Qorvo, 2020

AI Bin Smart Recycling Assistant @ Brisbane Airport

- Features & experiences: [Oscar Sort: Zero-Touch Zero-Waste Stations — Intuitive AI]
- YouTube video: [Oscar Sort, Increasing Smiles & Sustainability]
- BNE Airport's Sustainability Plan [Australian first for Brisbane Airport]



Smart Trolley at Woolworths Supermarket

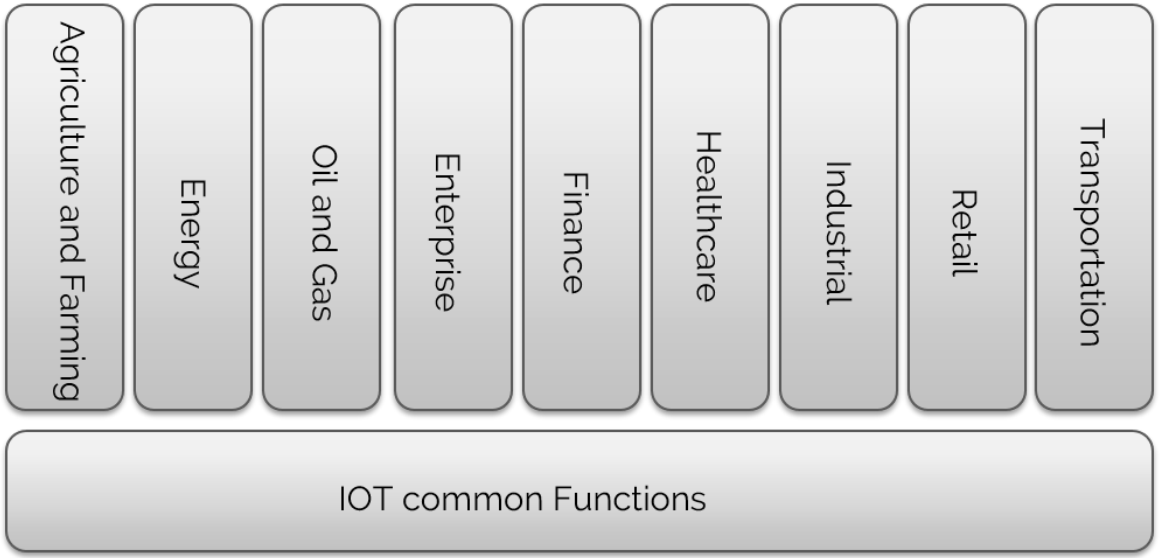
Your trolley knows what you spent before you checkout

- Scan every item – built in barcode scanner logs each product the moment it's added to the trolley.
- Real-time spend tracking – runtime total updates instantly, linking scan events to pricing & promotion data



Video link: Scan&Go Trolleys
Source: Woolworths Group, 2024

IoT Verticals by ETSI (European Telecommunications Standards Institute)

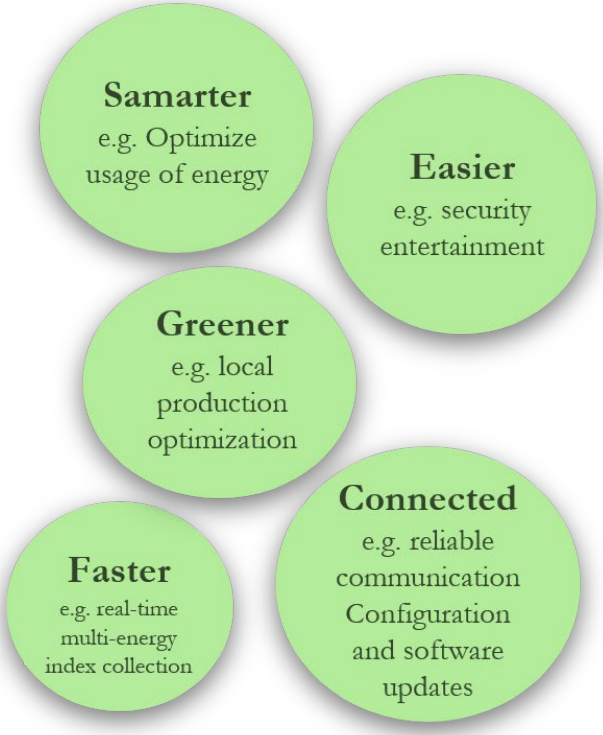


(Rayes and Salam, 2022)

IoT Vertical: Energy Sector

Energy

- Smart Meters
  - Real time monitoring of electrical power consumption
  - Advanced planning of energy supply
- Smart homes
  - Appliances monitoring
  - Security system
  - Motion sensor
  - Remote operation while reducing energy consumption



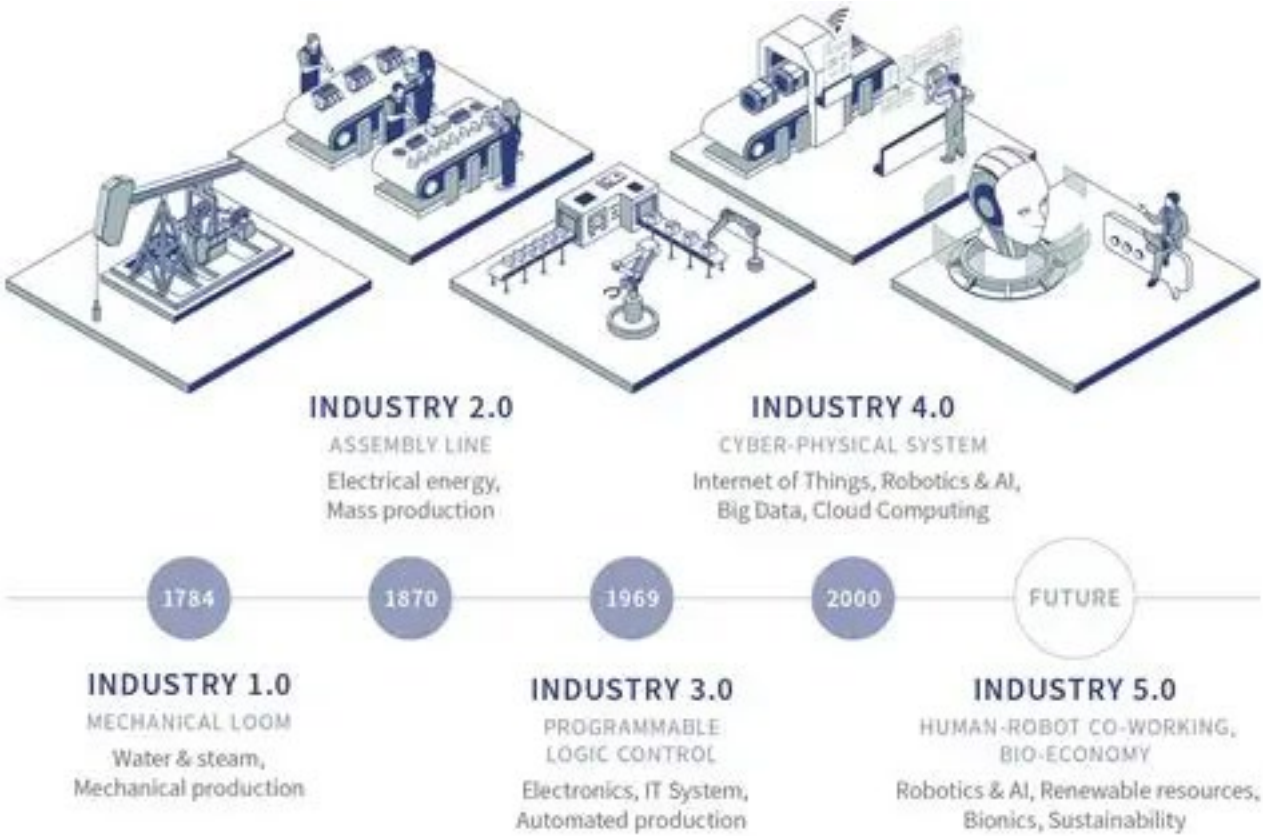
(Rayes and Salam, 2022)

IoT Applications: Smart City



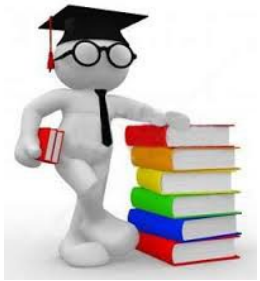
Industrial IoT (IIoT)

- A **key player** in Industry 4.0
- **Expand and improve analytics capability** by connecting billions of devices



Overview of the Unit



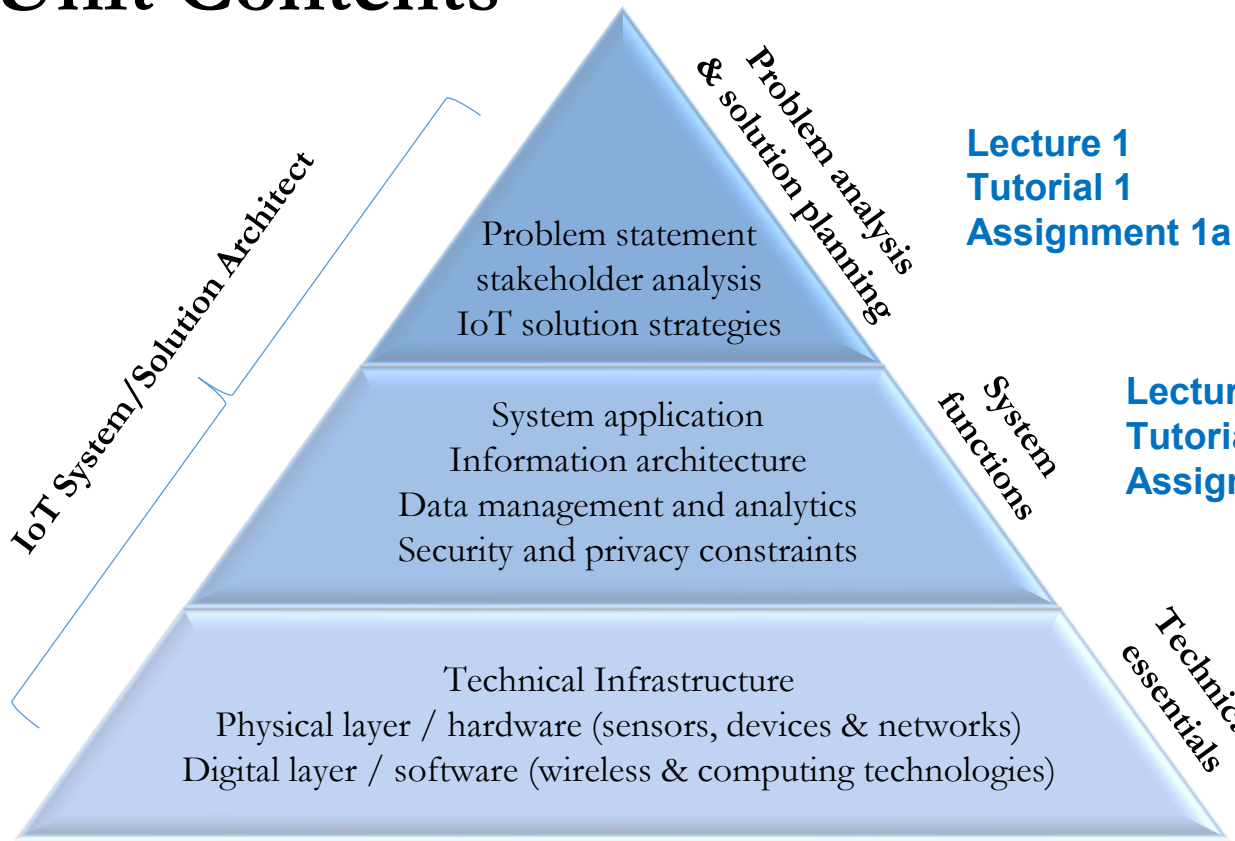




Unit Objectives

- The unit aims to deliver insights into IoT solution design and analysis (in the context of an **organisation** or **enterprise, i.e. Industry IoT**).
- Develop knowledge and skills in designing IoT solutions using
  - mobile and ubiquitous sensors and smart devices
  - network technologies and protocols
  - computing platforms
  - data management and analytics
  - strategies to address security and privacy considerations
- Develop skills in analysing, planning, and developing the design of functional and technical aspects for an IoT solution.

Unit Contents



Lecture 1
Tutorial 1
Assignment 1a

Lectures 5-6
Tutorial 5,
Assignment 2

Lectures 2-4
Tutorials 2-4
Assignment 1b

Unit Schedule

Date/Time	Monday (19/10)	Tuesday (20/10)	Wednesday (21/10)	Thursday (22/10)	Friday (23/10)
08:30 – 10:05	L1: Introduction & Unit Overview	L3: Network Technologies	L4: Fog/Edge and Cloud Computing	L5: Security and Privacy & T5: IoT System Security Analysis	**A1 Presentations (T1-12)**
10:25 – 11:50	L2: Sensors and Devices	T3: IoT Connectivity I	T4: IoT Connectivity II	A2 Guidance: IoT System Design	L6: Making Use of IoT Data
			Lunch Break		
13:50 – 14:35	T1: Solution Scoping	Work on A1a	Work on A1b	Work on A2	Unit Conclusion & Outlook
14:40 – 16:10	T2: Selecting IoT Devices	A1a review & feedback (interview)	A1b review & feedback (interview)	A2 review & feedback (interview)	**14:40 - 16:00** **Planning for QUT Final Year & Postgraduate Study**
16:15 – 16:30	Briefing A1a: IoT Solution Planning	Briefing A1b: IoT Solution Design			
16:35 – 17:20 *(Self-study)*	*Start A1a*	*Start A1b*	*Complete A1a & A1b*	*Prepare for A1 presentation*	

[Notes]

1. Abbreviations: Lecture (L), Tutorial (T), Assignment (A), Team (T)
2. Assignment Teams: 4 students per team and to register in T1 to T12
3. Individual assignment teams are interviewed to review progress and receive feedback
4. A1 presentation: 5 minutes presentation + 1 minute Q&A per team

Assessment Items

No.	Type	Description	Marks	Due date
1a	Assignment/Report	IoT Solution Planning	15	25 October
1b	Assignment/Report	IoT Solution Design	40	25 October
2	Assignment/Report	IoT System Design	30	1 November
1(ab)	Assignment/Presentation	IoT Solution Planning & Design	15	23 October

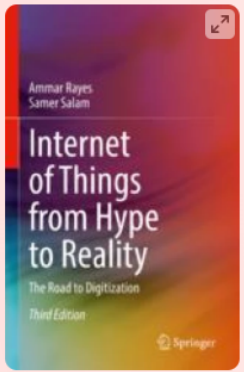
- **Although these are all team assignments, individual marks may vary based on individual contribution, progress review, and presentation performance.**

Key Reference

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The Road to Digitization

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